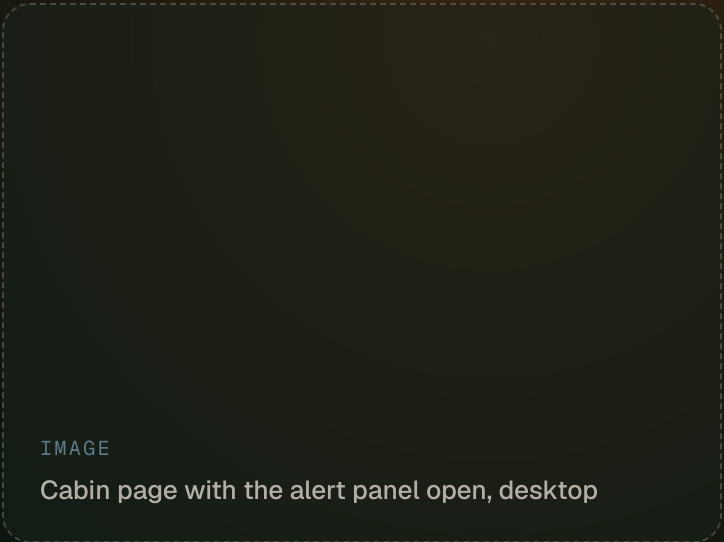


Refresh less, camp more.

Ember watches Recreation.gov for cancellations at Forest Service cabins and emails you when your dates open up. I designed and built it on my own, from the brand to the database.



IMAGE

Cabin page with the alert panel open, desktop

ROLE

Design and engineering, solo

TIMELINE

May 2026 to present

STACK

Next.js, Supabase, Resend, Mapbox, Vercel

LIVE

ember.gayatrisabne.dev

THE PROBLEM

Forest Service cabins book out months ahead. Plans change and cancellations happen constantly, but an opening is gone within minutes, taken by whoever happened to be refreshing the page.

WHO IT'S FOR

A local adventurer who already found the cabin they want and hit a wall. They'd go this weekend if something opened. Finding the cabin isn't the problem; the dead end after finding it is.

THE KEY DECISION

Don't compete with Recreation.gov. It owns the data and the brand recognition. Ember is a layer on top: the cabin page shows just enough to confirm the place, then gets out of the way of availability and alerts.

One warm light source

The name comes from the product's core moment: a cabin opening up after months dark. An ember is what's left when the fire has almost gone out. It's still glowing, and still an opportunity if you move fast.

The product lives at night, on your phone in bed when the alert arrives, so the visual language does too. Every competitor defaults to daylight. On near-black forest green, the same orange family stops being a brand color and becomes a light source, reserved for the most important element on each screen.



NIGHT	EVERGREEN	EMBER	SMOKE	WAX
#0F1510	#1A241B	#D45A20	#5F7A8A	#EDE8DC

Trail Creek Cabin

FRAUNCES · DISPLAY ONLY

Alert me when 2 nights open up

GEIST · INTERFACE

SLEEPS 4 · \$45 / NIGHT

GEIST MONO · DATA

Figma → code → Figma

Figma and the codebase are one system kept in sync both ways. Code isn't the end of a handoff; the build gets written back into the design file.

FIGMA

Variables and styles

Color variables, text styles and a four-step spacing rhythm (bonded, grouped, section, major) with mobile and desktop modes.

CODE

Tokens, locked

`theme.css` and `utilities.css` hold the same names. Components in `components/ui` use only tokens: no raw hex, no arbitrary px.



BACK TO FIGMA

The build, written back

13 `Ember/*` text styles generated from `utilities.css`, each stamped with its class name. Live screens rebuilt as audit frames.

LIVE

ember.gayatrisabne.dev

Deployed on every push to main. The real product, at real widths, with real data.



DRIFT IT CAUGHT

Writing styles back from code exposed two older Figma type ramps that had quietly diverged from the build, one of them mixed into the landing frames.

BUGS IT CAUGHT

Auditing the explore page at 393px found the search field squeezed to 120px and the mobile tab bar missing. Neither was visible in the desktop design.

Right-size the engineering to the data

Ember has two kinds of data with opposite shapes. That asymmetry, not a framework preference, decided the architecture: cheap data is loaded eagerly, and the expensive data only gets heavy machinery where it earns it.

	CABIN FACTS	AVAILABILITY
SHAPE	505 cabins from a one-time data dump. Small (about 30 KB gzipped) and frozen.	Live , per cabin and per month. Only Recreation.gov has it.
SO	The whole set ships to the map once. Panning makes zero cabin requests. The page is cached and rebuilt daily.	Fetches on demand when you open a cabin. Showing it across the whole map would take about 500 live calls, so that's deferred.
COMPARE	Airbnb fetches per viewport because it has millions of listings. At 505, loading everything is simpler and feels faster.	Alerts are the one place that has to read live availability on a schedule. That's the pipeline below.

THE ALERT CHECK

1 · TRIGGER

Scheduled cron

Vercel runs the check daily and authenticates with a shared secret, so nobody else can fire it.

2 · CHECK

Find openings

Groups alerts by cabin, fetches each month once, then finds runs of at least N open nights in each alert's dates. Reads only.

3 · SEND

Claim, then email

Claims a row guarded by a unique constraint, then emails through Resend. An opening already sent is skipped.

4 · REPORT

Count outcomes

Sent, duplicate and failed are tallied separately, so a broken write is never mistaken for a duplicate.

Ten watchers, one request

Fetches are batched per cabin, so popular cabins don't multiply load on Recreation.gov.

Safe to rerun

The database, not the code, guarantees each opening is emailed once. A retry or a manual run can't double-send.

Tested without a network

Check and send are separate modules that take their I/O as inputs, so both are unit-tested with fakes.

A KNOWN LIMIT

Checks run once a day because Vercel's free tier caps crons at daily. Nothing in the pipeline assumes that; on a paid plan it would run every 15 minutes. A dev trigger forces a run for demos.

Three calls about structure

BROWSE · MAP FIRST

A full-screen map, not a split list

Airbnb's 50/50 split works when there are 200 results in a few blocks and you need a list to compare. Cabins on public land are sparse. People orient by place first: which trails, is there a lake. So the map takes the whole screen and the cabin card floats on it, a move borrowed from AllTrails.

It's also the one light surface in a dark product. Light is for scanning a landscape; dark is for focusing on a decision.

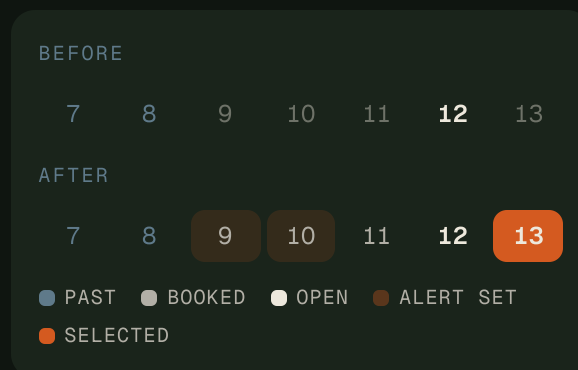
IMAGE

Explore page, full-screen map with a cabin card open

CALENDAR · CELL STATES

Availability was shown by dimming

Most nights at a popular cabin are booked, so most cells rendered dim and the calendar looked dead. Worse, booked nights you could tap were dimmer than past days you couldn't.



I split it into two channels. Brightness means availability, and only the past is truly dim. Ember means your alert or your selection. No borders anywhere.

ALERTS · MINIMUM NIGHTS

A toggle that couldn't say what people meant

BEFORE

Strict ± 7 days

AFTER

Alert me when at least 2 nights ▼

open up in your date range.

People wanted "any night this fall would do." Strict required the whole range to open, which never happens for a broad window. I tried a stepper, a segmented control and a sentence in Figma. The sentence won: it explains the model as you read it, and it scales from 1 night to 90.

The schema change shipped expand, migrate, contract, so the live app never ran against a database it didn't understand.

WHAT'S NEXT

Describe a trip, not a cabin. A national forest, drive-in, sleeps four, any weekend in July, and Ember watches everything that matches.

Book it for them. The hardest version, and the one that makes Ember feel like a friend watching Recreation.gov for you.

Smaller gaps. A way back from alert setup to the calendar, and one digest email when a broad alert finds many openings.